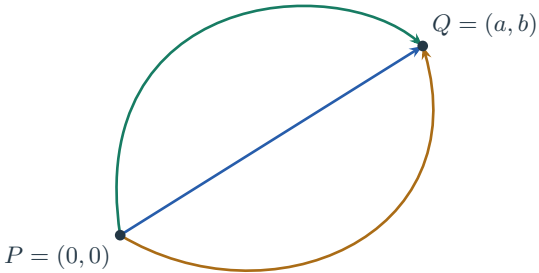
例：梯度场的线积分只由端点决定



三条路径的起点、终点和积分方向相同

$$\phi = x^2 + y^2,\ \nabla\phi = (2x, 2y)$$

$$\int_P^Q \nabla\phi \cdot d\mathbf{l} = \phi(Q) - \phi(P)$$

$$\phi(Q) - \phi(P) = a^2 + b^2$$

光滑路径改变时积分值保持不变。